LoRA versus full fine-tuning

MQP | Preliminary FP32 results | Prepared October 6, 2026

Purpose and comparison

Measure whether LoRA can learn a new task while training fewer parameters and using less GPU memory. We compare an unchanged pretrained reference, full fine-tuning (all original weights trainable), and LoRA fine-tuning (only added adapters trainable). Both trained methods receive exactly the same new data.

Experiment configuration

Pretrained GPT-2 in minGPT: 12 layers, 768 hidden units, 12 attention heads. Separate input and output embedding weights give 163,037,184 base parameters. Synthetic basketball data: 480 training matches / 1,440 questions; 40 validation matches / 120 questions; 40 test matches / 120 questions. Questions cover winners, halftime leaders, and comebacks.

Matched training protocol

NVIDIA A30; FP32; TF32 disabled. Batch size 8, padded sequence length 128, 1,200 steps, seeds 42/43/44. AdamW, zero weight decay, gradient clipping 1.0. LoRA rank 4, alpha 8, applied to attention and MLP projections. Starting weights, examples, sample order, and token budgets match between methods.

Model selection

Each method tried learning rates 0.001 and 0.0001 with seed 42. Validation loss selected full fine-tuning at 0.0001 and LoRA at 0.001; the chosen rates were then used for seeds 43 and 44. All selected checkpoints were at step 1,200. Test evaluation used identical prompts and greedy generation, capped at 40 new tokens.

Parameter counts

Measurement	Full fine-tuning	LoRA
Total parameters	163,037,184	163,627,008
Trainable parameters	163,037,184	589,824
Trainable fraction	100%	0.3605%
Changed scalar values (seed 42)	123,851,224	589,824
Frozen base values changed	Not applicable	0

How the frozen-weight claim was verified

Parameters were compared with cloned CPU snapshots of their initial values. Exact scalar comparisons distinguish values that actually changed from those merely allowed to change. All LoRA base-weight audits passed: zero original weight values changed. LoRA has 276.4 times fewer trainable parameters.

Source: saved matched-run reports, checkpoints, and familiar-name diagnostic.

Memory and training time

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Measured resource results

Measurement	Full fine-tuning	LoRA
FP32 parameter storage	621.94 MiB	624.19 MiB
Peak allocated GPU memory	3,747.0 MiB	2,481.6 MiB
Peak reserved GPU memory	4,060.0 MiB	2,758.0 MiB
Mean training-step time	149.8 seconds	105.2 seconds

Interpretation

LoRA reduced peak allocated training GPU memory by 33.8% and synchronized training-step time by 29.8%. Weight storage increased by 2.25 MiB because adapters are additional parameters. LoRA saves training resources while retaining the full pretrained model.

Storage is different from runtime memory

FP32 parameter storage equals parameter count multiplied by 4 bytes (32 bits). Peak allocated training memory also reflects live gradients, optimizer states, activations, and other PyTorch allocations. Reserved memory includes allocator-held space; it must not be added to allocated memory. These are PyTorch measurements, not total device memory including every driver allocation.

Timing scope

Training-step time is averaged over three seeds and covers the 1,200 synchronized training steps. It excludes data transfer, validation, parameter audits, checkpoint saving, and test generation. These results compare a fixed training budget; they do not establish time required to reach a particular quality target.

Baseline controls

FP16/BF16, FlashAttention, and KV caching were not introduced into this comparison. Keeping the experiment in FP32 isolates LoRA as requested. Combined-technique experiments and broader latency/throughput scaling remain future work.

Source: saved matched-run reports, checkpoints, and familiar-name diagnostic.

Quality, examples, and conclusions

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Original held-out test

Metric	Pretrained	Full fine-tuning	LoRA
Correct primary answer	25.8%	31.9 +/- 14.4%	81.7 +/- 0.8%
Answer and score fields	0.0%	31.9 +/- 14.4%	80.8 +/- 1.4%
Both score-swap answers correct	0.0%	6.4 +/- 7.7%	65.8 +/- 0.8%

What this test measures

Test team names and match-score combinations were absent from training, and none of the full reference answers appeared in training. Score-swapped questions test whether answers follow changed inputs. Reported +/- values are sample standard deviations across three training seeds, not confidence intervals. The benchmark was previously disclosed, not newly blinded.

Actual held-out example (seed 42)

Prompt: Bulls vs Sharks; halftime 51-35; final 96-69. Who won? Full fine-tuning: "Rockets won 96-69." LoRA: "Bulls won 96-69." The pretrained model generated contradictory winner statements. Full fine-tuning often substituted names seen during training.

Exploratory familiar-name diagnostic

After observing those failures, we replaced only test team names with familiar names, keeping scores and checkpoints unchanged. Primary-answer accuracy: pretrained 30.8%; full fine-tuning 82.5 +/- 2.5%; LoRA 82.5 +/- 0.0%. Both trained models correctly answered "Bears won 96-69" for the corresponding Bears-Hawks example. This diagnostic involved no retraining, but 9 of 120 reference answers overlap training.

Limits on the quality claim

This is synthetic, template-based basketball question answering, not broad sports conversation. Scoring uses format-sensitive answer and score-field checks. Questions share matches, and there are only three training seeds. Both methods achieved roughly 50% on comeback questions. The familiar-name diagnostic is exploratory and does not prove statistical equivalence; the unseen-name result does not establish that LoRA universally outperforms full fine-tuning.

Finding to present

Both methods learned the task with familiar team names. LoRA made only 0.3605% of its parameters trainable, preserved every base weight, and used less measured GPU memory and training-step time. It also generalized better to unfamiliar team names in this experiment. These are preliminary results supporting parameter efficiency, with clear remaining quality limitations.

Source: saved matched-run reports, checkpoints, and familiar-name diagnostic.